what is going to sustain," advises Raja.

"What I would like to see on a personal level is that a lot of product innovation happens in India. End product innovation and end device development needs to happen in India too. Because a lot of start-ups today focus more on applications and software, and this focus has to now change into devices, which makes more sense to India. Given the kind of chip manufacturing capability that we have we can do even more innovation. Many of the design houses here can make for India. Made in India and Make in India are great slogans but I would also like to see Design for India as a very key aspect coming in and government driving in more incentives and policy in that front," shares Balaji.

Saurabh adds, "I liked what Balaji said in the sense that if Indian companies are able to grow at scale and engage with these global companies, there is a possibility that in the long term they might be acquired. Even though there is a motive to keep the IP domestic but we would still want more Indian innovation to feed into global companies and hopefully see Indian companies also become global champions in this process."

Support from academia

Academia will also have to play an important role in enabling India as the hub of semiconductor and display manufacturing. Such companies originating in the likes of Taiwan, China, Korea, and Japan owe a lot of their success to academia, and the scenario in India, as Raja explains, will be no different. Raja further explains that as larger companies avoid taking risks, MSMEs might be the right ones to extend their capabilities and try hands at researching with the right institute(s). One such research agency has already existed at IIT Kanpur for more than 15 years and it is known as the National Centre for Flexible Electronics.

"Whether you look at TSMC, or any other such company, the academia has played a very important role in their success. The key for companies will be to find out how they can collaborate with academia and fine tune relationship building," shares Sudheer Kumar, COO, NCFlexE, IIT Kanpur.

Satya adds, "I think academia has a very big role to play here by having more MS and PhD programmes targeted at semiconductors. Going back to the success of Silicon Valley, one of the key reasons of their success was how the two big universities, Stanford and Berkeley, worked with the industry and then had the right skills and the right kind of innovation. It was done in collaboration between industry and the academia, so I urge all the academia to do the same thing and drive the innovation spirit."

He adds, "I think how MNCs like Intel, Samsung, and Qualcomm create alliances in India is also an interesting piece. We have to start collaborating with education institutions and academia more. So, I believe we need to have more programmes on chip manufacturing, because silicon design is complex. But the great news is that we have a good base to start with and we can only get better from here."

"To make embedded systems and VLSI more attractive, I think there needs to be more core work with Academia because there's more to it. AI/ML is important but it's not everything. Unless you get your basic VLSI architecture right, these things won't work because they operate at a much higher level. So, this would mean that a lot of collaboration from the industry is required," says Balaji.

The first-mover advantage

The proverb 'The early bird catches the worm' will be of much relevance to companies deciding to establish semiconductor and display fabs in the country. As explained earlier, there is a huge market waiting to be tapped and then there is the scope of possible exports as the world has started looking for alternatives to China. India is already a market of ten fabs and there exists none in the country.

"There is a first-mover advantage because primarily this is a capital-intensive industry. It is not easy to make an entry, unless there is good support available from the government. Whoever comes forward with initial two-three fabs in each of the sectors will have an advantage. However, what is also true is one or two fabs in the country are not going to satisfy the domestic market. The early movers will have a better learning curve in comparison to others as it is a very long-term game," says Pranav.

India, as suggested by most of the leaders, is at once-in-a-generation inflection point. The country has a huge market, a market that is only starting to mature. The next three decades are forecast to be centered around India. Corning's Sudhir is of the view that good implementation of already announced programmes and a consistency of the same will make India more conducive for conglomerates.

"The devil is in the execution. That is what will keep the market attractive for us," says Sudhir. EFY

Siddha is a business journalist who is fond of recognising and presenting how tech is shaping up the business world. When not writing about EVs and government policies, she enjoys reading non-fiction for fun. Mukul is passionate about how technology can change lives for good, and skeptical about how the same technology can make matters complicated for the very existence of human beings.

This article was developed thanks to sessions held by ELCINA in collaboration with EFY titled "Establishing an Eco-System For Semiconductor & Display Products in India," and by Society of VLSI in collaboration with Times Techies titled "Techade For Semiconductors - The Road Ahead."